ANDRES LEBBOS

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RESEARCH INTERESTS

Computational Oncology

Clinical Decision Systems

Stochastic Decision Making

Medical Image Analysis

Multi-Agent Systems

Network Dynamics

LANGUAGES

Spanish	*Native*	C2
English	*Fluent*	C2
Arabic	*Advanced*	C1
Italian	*Moderate*	B1
Netherlands	*Basic*	A2

PROFILE

AI researcher with 5 years of experience in building computational systems for data-driven applications, a background in Computer Science and Electrical & Computer Engineering and a Master's in Artificial Intelligence. Focused on cancer research and optimal treatment strategies for clinical decision support systems. Experienced in multi-agent systems, game theory, network dynamics, and computer vision for modeling complex biomedical interactions.

EDUCATION

Maastricht University — **Aug, 2024 - current**

GPA **9.09 / 10** — Maastricht, Netherlands

MSc Artificial Intelligence

Quantum Computing specialization

Duke University — **June 2016 - May 2019**

GPA **3.585 / 4** — Durham, NC

BSE in Electrical & Computer Engineering

BSc in Computer Science

San Carlos School — **Aug. 2003 - June 2015**

GPA **3.8 / 4** — Bogota DC, Colombia

High School, Middle School, and Elementary School Diploma

WORK EXPERIENCE

Software Reliability Engineer — **June 2022 - Feb. 2024**

Art of Problem Solving — Bogota DC, Colombia

Maintained, monitored, and modernized the reliability, resilience, infrastructure, and security of over a dozen different products, reaching over a million users.

Organized, advised, and executed a committee, reducing product degrading incidents by 20% and improving the approach and performance of their resolution and prevention.

Software Engineer — **Sep. 2019 - June 2022**

Art of Problem Solving — San Diego, CA

Designed and implemented dozens of software tools in an online school to improve the educational experience for 10,000 students, 270,000 members, and the teaching staff.

Coordinated and directed interview segments for the final hiring of more than two dozen software engineers.

Advised and guided new software engineers with the onboarding process to familiarize them with the distinct infrastructures, technologies, and procedures used for development.

TECH SKILLS

Programming Languages

Python Java

MATLAB C

Research Infrastructure

LaTeX Unix

Hugging Face GitHub

Machine Learning & Data

PyTorch TensorFlow

NumPy SQL

Software & Systems

AWS Docker

COURSEWORK

Design & Analysis Algorithms

Advanced Machine Learning

Advanced NLP

Computer Vision

Multi-Agent Systems

Dynamic Game Theory

Stochastic Decision Making

Random Noise & Signals

Autonomous Robotic Systems

Modern Control Systems

Quantum Algorithms

Quantum AI

PROJECTS

Big Data Migration **July 2022 - Oct. 2023**
Art of Problem Solving Durham, NC

Migrated and optimized a 1TB database, resulting in a 20% performance increase and a 40% cost reduction without any data loss.

Automatic Student Grader **Jan. 2021 - Jan. 2022**
Art of Problem Solving Durham, NC

Developed an algorithm for automatic student grading, achieving a 97% success rate while effectively flagging academic dishonesty.

Sonar Bathymetry Hardware **Jan. - May 2019**
Duke University Durham, NC

Constructed an economic robot capable of diving up to 10,000 feet to track marine species and map oil reserves for research purposes.

Multiple Robot Synchronized Path Planning **Aug. - Dec. 2018**
Duke University Durham, NC

Designed an efficient, collision-free path-planning algorithm for dozens of synchronized robots in tight passages.

Vehicle Slippage in Unknown Terrain **Aug. - Dec. 2018**
Duke University Durham, NC

Created a learning algorithm for four-wheel steering vehicles, reducing slippage in unknown terrains at high velocities.

Measured the coefficient of friction in an unknown terrain through machine learning, feedback, and PID controllers.

Role-Playing Game (RPG) Engine **Jan. - May. 2017**
Duke University Durham, NC

Created an RPG Engine allowing users with no coding experience to design games for the final creation of dozens of innovative ideas.

Designed, implemented, and tested the back-end software, supporting the simultaneous usage by massive amounts of players.

Self-Sustained Farming Project **Aug. - Dec. 2015**
Universidad de los Andes Bogota DC, Colombia

Designed and constructed a cheap, award-winning, self-sustained automatic farming environment to help Colombian farmers.

AWARDS

Ibero-American Mathematical Olympiad **2013**
Bronze Medal Panama City, Panama

Central American & Caribbean Mathematical Olympiad **2013**
Bronze Medal Managua, Nicaragua